Ruo-Yu Guan

Curriculum vitae · October 2026
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Research Profile

- Galactic double white dwarfs and Galactic structure inference through population synthesis and Bayesian methods (LISA / TianQin).
- Residual Galactic binary foregrounds and stochastic gravitational-wave background inference.
- Earlier work: gamma-ray burst prompt emission, polarization modelling, time-series analysis, and multiwavelength source identification (Chandra / Gaia).

Education

Huazhong University of Science and Technology — Wuhan, China
School of Physics; National Gravitation Laboratory — *Sep. 2025 – Jun. 2029 (expected)*
Ph.D. in Physics (Theoretical Physics, in progress) — *Supervisor: Prof. Yan Wang*

- Research focus: gravitational-wave astrophysics (Galactic double white dwarfs).

The University of Hong Kong — Hong Kong SAR, China
Department of Physics, Faculty of Science — *Nov. 2024*
M.Sc. in Physics — *Supervisor: Assoc. Prof. Stephen Chi-Yung Ng*

- Capstone project: *Identifying Sources in Cygnus OB2 Using Multiwavelength Observations*.

Jilin University — Changchun, China
B.Sc. — *Jun. 2022*

- Graduation thesis: *Study on the Prompt Emission of Gamma-ray Bursts and its Polarization*.
- Research mentor: Prof. Mi-Xiang Lan (Center for Theoretical Physics & College of Physics, Jilin University).

Research Positions

Huazhong University of Science and Technology — Wuhan, China
Research Assistant (contract), Department of Astronomy — *Nov. 2024 – Jun. 2025*

- Supervisor: Prof. Yuan-Chuan Zou.
- Gamma-ray burst research (see Research Experience / Publications).

Publications

Preprints

1. **Ruo-Yu Guan** and Yan Wang (2026).
   "Residual Galactic binary foreground in LISA stochastic gravitational-wave background inference: source power concentration and spectral degeneracy."
   arXiv:2607.25349 [astro-ph.HE].
   arXiv

Refereed Journal Articles

1. **Ruo-Yu Guan**, Fei-Fei Wang, and Yuan-Chuan Zou (2026).
   "Hurst index of gamma-ray burst light curves and its statistical study."
   Journal of High Energy Astrophysics, **51**, 100559.
   DOI: 10.1016/j.jheap.2026.100559

2. **Ruo-Yu Guan** and Mi-Xiang Lan (2023).
   "Interpreting time-integrated polarization data of gamma-ray burst prompt emission."
   Astronomy & Astrophysics, **670**, A160.
   DOI: 10.1051/0004-6361/202243805

Research Experience

Project dates below describe the period of research activity; degree and employment dates are listed separately above.

Galactic Double White Dwarfs and Space-Based Gravitational-Wave Inference Jul. 2025 – Present
National Gravitation Laboratory and School of Physics, Huazhong University of Science and Technology *Wuhan, China*
Research with Prof. Yan Wang; Ph.D. enrolment from Sep. 2025

- Research focus: gravitational-wave astrophysics and Milky Way structure inference using Galactic double white dwarf binaries.
- Building a simulation-based inference pipeline integrating population synthesis (COSMIC) with hierarchical Bayesian inference (GWpopulation).
- Quantifying selection effects, detector response, and observational uncertainties for space-based gravitational-wave observations (Laser Interferometer Space Antenna / TianQin context).
- First-author preprint on residual Galactic binary foregrounds (arXiv:2607.25349, with Prof. Yan Wang); presented at the 2026 Mid-Year Workshop in Yangzhou.

Time-Series and Correlation Analysis of Gamma-Ray Burst Light Curves Oct. 2024 – Jan. 2026
Department of Astronomy, Huazhong University of Science and Technology *Wuhan, China*
Research with Prof. Yuan-Chuan Zou; Research Assistant appointment Nov. 2024 – Jun. 2025

- Conducted gamma-ray burst light-curve analysis using detrended fluctuation analysis and Hurst exponent estimation; performed statistical correlation studies in Python.
- Maintained Python-based workflows for preprocessing, feature extraction, statistical analysis, and visualization.
- Resulted in a first-author refereed journal article.
- Supported by the National Square Kilometre Array Program of China (Grant No. 2022SKA0130100).

Identifying Sources in Cygnus OB2 Using Multiwavelength Observations Sep. 2023 – Jun. 2024
Department of Physics, Faculty of Science, The University of Hong Kong *Hong Kong SAR, China*
M.Sc. capstone project (Supervisor: Assoc. Prof. Stephen Chi-Yung Ng)

- Reduced and analyzed *Chandra* X-ray observations using the Chandra Interactive Analysis of Observations software.
- Merged 18 X-ray observations in the Cygnus OB2 region and assembled a catalog of detected X-ray sources.
- Cross-matched X-ray sources with *Gaia* counterparts to classify association members versus foreground/background objects.

Gamma-Ray Burst Prompt Emission and Polarization 2022 – Feb. 2023

Center for Theoretical Physics and College of Physics, Jilin University *Changchun, China*

Research with Prof. Mi-Xiang Lan, originating in the B.Sc. thesis project

- Collected and organized spectral parameters and polarization measurements of gamma-ray burst prompt emission from the literature.
- Computed time-integrated polarization degrees for ~37 gamma-ray bursts under a synchrotron-emission framework using Interactive Data Language.
- Resulted in a first-author refereed journal article.
- Supported by the National Natural Science Foundation of China (Grant Nos. 11903014 and 12147217).

Skills

- **Programming:** `Python` (primary); `MATLAB`, `Mathematica`, `R` (familiar); `IDL`, `C++` (prior experience).
- **Methods:** time-series analysis; statistical correlation analysis; hierarchical Bayesian inference (`GWpopulation`; Markov chain Monte Carlo); Monte Carlo simulation and mock-catalog generation.
- **Scientific computing:** `NumPy`, `SciPy`, `pandas`, `Matplotlib`; `Jupyter`; Flexible Image Transport System data handling; pipeline automation (`bash`/`Python` scripts).
- **Astrophysics & gravitational-wave:** population synthesis (`COSMIC`); gravitational-wave population inference; catalog-based population studies.
- **Scholarly writing:** `LaTeX`; `BibTeX`.
- **Computing environment:** `Git`/`GitHub`; `bash` shell scripting; Windows, macOS, Linux (Ubuntu); high-performance computing (remote Linux servers, batch jobs); `SSH` workflows.

Software and Open Science

- **Scientific Textbook Proofreading Skill** — Open-source pre-release for structured scientific textbook proofreading and errata auditing (May 2026). Includes report templates, a JSON schema, synthetic examples, evaluation cases, and CI checks; produces candidate errata for human expert review.

Talks and Presentations

2026 Mid-Year Workshop of the National Key R&D Program Project on Space Gravitational-Wave Detection Signal Simulation, Processing, and Verification Jul. 2026

Oral presentation: "Galactic Binary Residuals in LISA SGWB Amplitude Inference: Source Power Concentration and Spectral Degeneracy"

Yangzhou, Jiangsu, China · Talk and slides

2024 PKU International PhD Student Forum on the Frontiers of Modern Astronomy Dec. 2024

Poster presentation: "Time-integrated Polarizations of Gamma-ray Burst Prompt Phase"

Kavli Institute for Astronomy and Astrophysics, Peking University · Poster

Additional Conferences and Schools

- Aug. 2025 — Participant, Gravitational Wave Data Analysis Summer School (Lanzhou Center for Theoretical Physics)
- Jun. 2025 — Participant, The First Edinburgh School for Extragalactic Astronomy (Institute for Astronomy, The University of Edinburgh)
- May 2025 — Participant, Gravitational Wave Open Data Workshop 2025 (Gravitational Wave Open Science Center)
- Dec. 2024 — Volunteer (conference support), International Conference on Space Sustainability 2024
- Jul. 2024 — Participant, Celestial Holography Summer School 2024 (Perimeter Institute for Theoretical Physics)
- Nov. 2023 — Participant, International Symposium on Cosmology and Particle Astrophysics 2023
- Nov. 2023 — Participant, Multimessenger Astronomy: Bridging Transients, Lensing, and Dark Matter (Cosmic Frontiers)

Professional Activities

- Mar. 2026 – Present, Core Member (Astrophysics Working Group) — Laser Interferometer Space Antenna (LISA) Consortium
- Jan. 2026 – Feb. 2026, Community Member — Laser Interferometer Space Antenna (LISA) Consortium
- Oct. 2023 – Nov. 2024, Member — Laboratory for Space Research, The University of Hong Kong
- May 2021 – Apr. 2022, Student Member — Chinese Physical Society

Outreach Activities

- Jan. 2025 – Feb. 2025, Participant — China Skywatcher Asteroid Search Campaign
  International Astronomical Search Collaboration